

Mohamad Halal

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SKILLS

Programming: Java, JavaScript, TypeScript, Python, C#, C, C++, SQL, Bash/Shell

Full-Stack & Integration: React, REST APIs, JSON, API integration with third-party services, client-server architecture, data flows between systems, modular service design

Databases: SQL, SQLite, relational databases, schema design, query design

Tools & Practices: Git/GitHub (branching, rebasing, code reviews), Jira, CI/CD, Docker, JUnit, Agile/Scrum, AI-assisted development (Claude Code)

Testing & Quality: Unit testing (JUnit, Python unittest), integration testing, regression testing, automated testing, debugging, root-cause analysis

PROJECTS

MyTravelPal — API Integration and React Frontend

2025

Third-Party API Integration, REST APIs, and Data-Driven UI — Personal Project

- Built a React/JavaScript web application that integrates with multiple third-party REST APIs, as measured by a live-deployed app fetching and displaying real-time restaurant and attraction data, by consuming the Google Maps and Places APIs and rendering location-aware results in an interactive UI.
- Worked with APIs and the data flowing between external services, as measured by correctly parsed and displayed results across varying API response formats, by handling JSON data mapping, error states, and graceful fallbacks when external services returned unexpected data.

CyberFinance — Full-Stack Portfolio Management System

Dec 2025 – Jan 2026

Full-Stack Development, REST API Design, and Integration Testing — Personal Project

- Designed and built a full-stack application with a C backend, a SQLite relational database, and a React/TypeScript frontend, as measured by 11 working REST API endpoints serving a browser dashboard, by understanding the data problem each endpoint solves and separating server, database, and UI layers behind defined API contracts.
- Implemented session-based authentication and per-user data isolation, as measured by zero cross-account data leakage across all test scenarios, by scoping every endpoint to the authenticated session and enforcing access control in the database layer.
- Wrote and ran automated tests covering functional behaviour, authentication, and concurrency before shipping, as measured by a repeatable validation cycle, by writing targeted test cases and performing root-cause analysis to debug defects across multiple integrated layers.

LRZIP Windows Port

May 2026 – Sep 2026

Large Codebase, AI-Assisted Development, and Careful Code Review — Personal Project

- Delivered the first native Windows port of LRZIP, a large POSIX-oriented C/C++ codebase, as measured by working builds on a platform never previously supported, by analyzing an unfamiliar architecture and adapting low-level OS-specific interfaces.
- Used AI tools (Claude Code) to accelerate code investigation, implementation, and testing, reviewing all generated output carefully and catching platform-specific issues before they shipped, as measured by verified correctness across the full codebase.
- Built an automated cross-platform validation workflow, as measured by output matching the Linux baseline byte-for-byte, by scripting repeatable build-and-test cycles and validating results across environments.

EXPERIENCE

Statistics Canada — Data Operations Centre (DOC), Census 2026

Gatineau, Quebec

Operator, Processing Operations and Census Help Line

May 2026 – Sep 2026

- Validated and processed high-volume data flowing between internal systems under strict quality and security requirements, as measured by throughput meeting production targets, by performing structured data validation, analytical review, and quality-control procedures.
- Investigated data discrepancies where getting the small details right was critical, as measured by timely resolution and escalation, by applying root-cause analysis and documenting findings for downstream teams.
- Collaborated across a large multi-disciplinary operation, as measured by clear cross-team coordination, by communicating technical findings to colleagues and end users and adapting to evolving priorities.

CERTIFICATES

QNX (BlackBerry) — Realtime Programming for the QNX OS

Ottawa, Ontario

Professional Training

Jan 2026 – Mar 2026

- Completed training in POSIX-compliant real-time systems programming, covering scheduling, inter-process communication, concurrency, processes/threads, and systems-level debugging.

EDUCATION

Carleton University

Ottawa, Ontario

B.Eng in Software Engineering — GPA: 3.9/4.0, Dean's List

Expected Graduation: April 2028